

Dosing Tanks

Anyone who works with chemicals, needs to store them safely. ProMinent dosing tanks are tough and ideal for working with metering pumps.

ProMinent®



PE storage tanks produced in a rotation process. ProMinent metering pumps, suction lances and stirrers can all be added. The stackable PE collecting pans are available in matching sizes.

ProMinent manufactures the robust dosing tanks in a rotation process.



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Dosing Tanks

Anyone who works with chemicals, needs to store them safely. ProMinent dosing tanks are tough and ideal for working with metering pumps.

Natural coloured/transparent PE metering tank

Threaded bush for the dosing pumps	Usable capacity l	Dim. D mm	Dim. H mm	Weight kg	Order no.
without threaded sockets	35	350	485	3.5	791993
gamma/ X, beta	60	410	590	5.0	791994
alpha, beta, gamma/ X	100	500	760	7.0	1001490
alpha, beta, gamma/ X	140	500	860	9.5	791995
alpha, beta, gamma/ X, Sigma/ 1/ 2/ 3, gamma/ XL	250	650	1,100	17.5	1023175
2 x gamma/ X, 2 x Sigma/ 1, 2, 3, 2 x gamma/ XL, 2 x beta	500	820	1,215	33.0	791997
2 x gamma/ X, 2 x Sigma/ 1, 2, 3, 2 x gamma/ XL, 2 x beta	1000	1,070	1,260	51.0	1010909
2 x gamma/ X, 2 x Sigma/ 1, 2, 3, 2 x gamma/ XL, 2 x beta	1500	1,150	1,735	80.0	1060975

Natural coloured/transparent PE metering tank

Designed for the installation of a manually operated or electric stirrer.

The 35 – 1,000-litre storage tank have an R 3/4" threaded sleeve (1,500 l: R 1 1/4") for drainage that can be drilled to Ø 10 mm on-site if required. A PE R 3/4" sealing stopper (1,500 l: R 1 1/4") with a seal is screwed in.

Dosing tanks without ProMinent logo are available on request.

Black PE metering tank

For light-sensitive media (UV-stabilised). Conductive material for discharging static electric discharges to protective earth conductor potential.

Usable capacity l	Order no.
35	1039183
60	1039184
100	1039185
140	1039186
250	1039187
500	1039188
1000	1039189
1500	1139861

Dosing Tanks

Anyone who works with chemicals, needs to store them safely. ProMinent dosing tanks are tough and ideal for working with metering pumps.

Natural/transparent PE dosing tank with flat mounting surface

Threaded bush for the dosing pumps	Dim. D	Dim. H	Usable capacity	Weight	Order no.
	mm	mm	l	kg	
without threaded sockets	410	590	60	5.0	1061060
without threaded sockets	500	760	100	7.0	1008599
without threaded sockets	650	1,100	250	17.5	1061061

Your benefits

- 'Natural/transparent PE dosing tank' design without sintered threaded socket
- Level mounting surface for the installation of metering pumps on the storage tank
- Additional installation of a manual or electric stirrer is possible

Made of UV-stabilised polyethylene in a stackable design with ProMinent logo. 2 flat sides for fixing the collecting pan.

PE colourless/transparent stackable collecting pans

Usable capacity	D1	D2	Dim. H	Weight	Order no.
l	mm	mm	mm	kg	
35	507	565	220	3.0	1010879
60	607	680	270	4.3	1010880
100	727	802	320	6.5	1010881
140	727	811	370	7.0	1010882
250	807	917	520	11.0	1010883
500	1009	1155	670	16.0	1010884

Black PE Collecting Pan

Usable capacity	D1	D2	Dim. H	Weight	Order no.
l	mm	mm	mm	kg	
35	507	565	220	3.0	1139862
60	607	680	270	4.3	1139863
100	727	802	320	6.5	1139974
140	727	811	370	7.0	1081705
250	807	917	520	11.0	1139975
500	1009	1155	670	16.0	1139976

An R 3/4" threaded sleeve is moulded on 35 – 500 litre collecting pans for drainage. This sleeve may require drilling (Ø 10 mm) on-site. An R 3/4" PE sealing stopper with a seal is screwed in (accessory part no. 200692).

Natural PE Collecting Pan

Usable capacity	D1	D2	Dim. H	Weight	Order no.
l	mm	mm	mm	kg	
1000	1200	1280	980	34.0	740719
1500	1350	1410	1,280	42.0	1060980

PE black stackable collecting pans

Usable capacity	D1	D2	Dim. H	Weight	Order no.
l	mm	mm	mm	kg	
1000	1200	1280	980	34.0	740726

Dosing Tanks

Anyone who works with chemicals, needs to store them safely. ProMinent dosing tanks are tough and ideal for working with metering pumps.

Depósitos de dosificación

Si se trabaja con sustancias químicas es necesario almacenarlas de forma segura. Los depósitos dosificadores de ProMinent son robustos y están perfectamente preparados para trabajar con bombas dosificadoras.

ProMinent®



Recipientes de PE fabricados por moldeo rotacional. Se pueden complementar con bombas dosificadoras, lanzas de aspiración y agitadores de ProMinent. Disponemos de bandejas colectoras apilables de PE en varios tamaños para cada necesidad.

Los robustos depósitos dosificadores de ProMinent se fabrican mediante el método de rotación.



Technical changes reserved. Printed in Germany, 6-9-2023.

Depósitos de dosificación

Si se trabaja con sustancias químicas es necesario almacenarlas de forma segura. Los depósitos dosificadores de ProMinent son robustos y están perfectamente preparados para trabajar con bombas dosificadoras.

Depósitos de almacenamiento de PE, de color natural/ transparente

Preparado para roscar las bombas	Volumen	Medida D	Medida H	Peso	N.º de referencia
	l	mm	mm	kg	
alpha, beta, gamma/ X, Sigma/ 1/ 2/ 3, gamma/ XL	250	650	1.100	17,5	1023175
2 x gamma/ X, 2 x Sigma/ 1, 2, 3, 2 x gamma/ XL, 2 x beta	500	820	1.215	33,0	791997
2 x gamma/ X, 2 x Sigma/ 1, 2, 3, 2 x gamma/ XL, 2 x beta	1000	1.070	1.260	51,0	1010909
2 x gamma/ X, 2 x Sigma/ 1, 2, 3, 2 x gamma/ XL, 2 x beta	1500	1.150	1.735	80,0	1060975
sin agujeros con rosca	35	350	485	3,5	791993
gamma/ X, beta	60	410	590	5,0	791994
alpha, beta, gamma/ X	100	500	760	7,0	1001490
alpha, beta, gamma/ X	140	500	860	9,5	791995

Depósitos de almacenamiento de PE, de color natural/ transparente

Preparados para la instalación de un agitador manual o eléctrico

Los recipientes de 35 – 1.000 litros disponen de un manguito con rosca R 3/4" para el vaciado (1.500 l: R 1 1/4") que en caso necesario el cliente puede perforar hasta un diámetro de 10 mm. Disponen de un tapón de cierre de PE R 3/4" (1.500 l: R 1 1/4") con junta.

También disponemos de depósitos dosificadores sin el logotipo de ProMinent a petición del cliente.

Depósitos de dosificación en PE negro

Para líquidos sensitivos a la luz

Volumen	N.º de referencia
l	
35	791998
60	791999
100	1001322
140	792000
250	1023176
500	792002
1000	1010912
1500	1060976

Depósitos de dosificación en PE azul

Volumen	N.º de referencia
l	
35	1003812
60	1003813
100	1003814
140	1003815
250	1023177
500	1003817
1000	1010913
1500	1060977

Depósitos de dosificación

Si se trabaja con sustancias químicas es necesario almacenarlas de forma segura. Los depósitos dosificadores de ProMinent son robustos y están perfectamente preparados para trabajar con bombas dosificadoras.

Depósitos de dosificación en PE amarillo

Volumen	N.º de referencia
I	
35	1003818
60	1003819
100	1003820
140	1003821
250	1023178
500	1003823
1000	1010914
1500	1060978

Depósitos de dosificación en PE rojo

Volumen	N.º de referencia
I	
35	1003824
60	1003825
100	1003826
140	1003827
250	1023179
500	1003829
1000	1010915
1500	1060979

Depósitos de dosificación

Si se trabaja con sustancias químicas es necesario almacenarlas de forma segura. Los depósitos dosificadores de ProMinent son robustos y están perfectamente preparados para trabajar con bombas dosificadoras.

Depósito dosificador PE natural/transparente con superficie de montaje plana

Preparado para roscar las bombas	Medida D	Medida H	Volumen	Peso	N.º de referencia
	mm	mm	l	kg	
sin agujeros con rosca	410	590	60	5,0	1061060
sin agujeros con rosca	500	760	100	7,0	1008599
sin agujeros con rosca	650	1.100	250	17,5	1061061

Ventajas clave

- Versión como el "Depósito dosificador PE natural/transparente", aunque sin conexión de rosca sinterizada
- Superficie de montaje plana para la instalación de bombas dosificadoras sobre el recipiente
- Posibilidad de instalar también un agitador manual o eléctrico

Fabricadas en polietileno estabilizado con UV en versión apilable con el logo ProMinent La bandeja colectora está diseñada con 2 aplanamientos laterales para facilitar la fijación.

Cubetos de retención sin color/transparentes de PE

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
35	507	565	220	3,0	1010879
60	607	680	270	4,3	1010880
100	727	802	320	6,5	1010881
140	727	811	370	7,0	1010882
250	807	917	520	11,0	1010883
500	1009	1155	670	16,0	1010884

Bandeja colectora PE negra

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
35	507	565	220	3,0	1010885
60	607	680	270	4,3	1010886
100	727	802	320	6,5	1010887
140	727	811	370	7,0	1010888
250	807	917	520	11,0	1010889
500	1009	1155	670	16,0	1010890

Cubetos de retención en PE azul

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
35	507	565	220	3,0	1010891
60	607	680	270	4,3	1010892
100	727	802	320	6,5	1010893
140	727	811	370	7,0	1010894
250	807	917	520	11,0	1010895
500	1009	1155	670	16,0	1010896

Depósitos de dosificación

Si se trabaja con sustancias químicas es necesario almacenarlas de forma segura. Los depósitos dosificadores de ProMinent son robustos y están perfectamente preparados para trabajar con bombas dosificadoras.

Cubetos de retención en PE amarillo

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
35	507	565	220	3,0	1010897
60	607	680	270	4,3	1010898
100	727	802	320	6,5	1010899
140	727	811	370	7,0	1010900
250	807	917	520	11,0	1010901
500	1009	1155	670	16,0	1010902

Cubetos de retención en PE rojo

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
35	507	565	220	3,0	1010903
60	607	680	270	4,3	1010904
100	727	802	320	6,5	1010905
140	727	811	370	7,0	1010906
250	807	917	520	11,0	1010907
500	1009	1155	670	16,0	1010908

Las bandejas colectoras de 35 – 500 l disponen de un manguito con rosca R 3/4" para el vaciado que en caso necesario el cliente puede perforar hasta un diámetro de 10 mm. Disponen de un tapón de cierre de PE R 3/4" con junta atornillado (accesorio n.º de referencia: 200692).

Bandeja colectora PE natural

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
1000	1200	1280	980	34,0	740719
1500	1350	1410	1.280	42,0	1060980

Bandeja colectora PE negra

Volumen	D1	D2	Medida H	Peso	N.º de referencia
l	mm	mm	mm	kg	
1000	1200	1280	980	34,0	740726
1500	1350	1410	1.280	42,0	1060981

Prominent Dosing Pump

CONCEPT c Solenoid Metering Pump

Capacity range : 0.7~23 l/h
Max. backpressure :16~1.5 bar
Drive mode : Solenoid-drive
Control type : manual,
external control

The ProMinent® CONCEPT c is a microprocessor-controlled, short-stroke, solenoid-driven, diaphragm-type metering pump for metering liquid, non-combustible media. It is popular to our customer for its precise design, reliable operation, good performance-cost ratio. It is widely applied to all kinds of chemicals dosing with a good anti-corrosion performance of the liquid end.



High lights:

- Solenoid-driven positive displacement metering pump, compact, reliable & durable.
- Capacity range from 0.7 l/h to 23 l/h, at a max. backpressure of 1.5 bar to 16 bar
- Multiple choices of liquid end materials for anti -corrosion purpose
- Stroke length is adjustable manually from 0 to 100%.
- Various stroke frequency control modes on request:
 - ✧ 3 constant metering frequencies manually: 100%, 50%, 25%
 - ✧ Potential free contact input & 50%, 100% constant frequencies.
 - ✧ 4-20mA input & 50%, 100% constant frequencies.

Liquid end material:

- PP1=PP/EPDM seal
- PP2=PP/FPM seal(Viton®)
- NP1=NP/FPM seal(Viton®)
- NP2=NP/EPDM seal
- *NP6=NP/FPM seal(Viton®)
- *NP7=NP/EPDM seal
- PVT=PVDF/FPM seal(Viton®)

* All options except for NP6 & NP7
with bleeding valves



▲ Injection valve, Foot valve,
Multifunction valve, Suction line

▼ Mounted on tank to make
dosing system.

General Data:

■ Metering deviations:	All material versions -5 % to +15 %
■ Metering reproducibility:	±2 %
■ Permissible ambient temperature:	-10 °C to +45 °C
■ Electrical connection:	220 V±10 %, 50 Hz (60 Hz is available for special requirement)
	110 V±10 %, 50 Hz (60 Hz is available for special requirement)
■ Type of enclosure:	IP 65
■ Insulation class:	F



Capacity Data

Type	Pressure	Max. capacity	Capacity	Stroke rate	Type	Pressure	Max. capacity	Capacity	Stroke rate	Type	Pressure	Max. capacity	Capacity	Stroke rate	Connection Inner/Outer	Permissible admission pressure	Weigh
	bar	l/h	ml/ stoke	strokes /min		bar	l/h	ml/ stoke	strokes /min		bar	l/h	ml/ stoke	strokes /min	mm	bar	kg
1600	16	0.7	0.125	90	1601	16	1.0	0.14	120	1602	16	1.5	0.140	180	6×4	8	2.5
1201	12	1.21	0.215	90	1202	12	1.7	0.24	120	1203	12	2.5	0.24	180	6×4	5.5	2.5
1001	10	1.63	0.453	60	1002	10	2.3	0.426	90	1003	10	3.4	0.472	120	8×5	3	2.7
0703	7	2.66	0.47	90	0803	8	3.4	0.48	120	0806	8	5.3	0.48	180	8x5	3	2.5
0306	3	6.08	1.08	90	0308	3	8.4	1.17	120	0313	3	13	1.20	180	8×5	1.5	2.7
0212	1.5	11.5	2.05	90	0215	1.5	14.8	2.05	120	0223	1.5	23	2.13	180	12×9	0.8	2.9

CONCEPT c | Bomba dosificadora

Operación sencilla, precisa y segura.



Imagen meramente ilustrativa.

ProMinent®

Rango de dosificación 0,72 – 24 L/h, 16 – 1,5 bar

La CONCEPT c ProMinent® es una bomba dosificadora de diafragma, controlada manualmente o remotamente, para la dosificación de líquidos no inflamables. Es reconocida por nuestros clientes por su diseño preciso, funcionamiento confiable y rentable. Es ampliamente aplicada a todos los tipos de productos químicos de dosificación con un buen desempeño anticorrosión del cabezal.

Ventajas

- Bomba dosificadora de desplazamiento positivo con accionamiento por solenoide, compacta, confiable y durable;
- La longitud de carrera es ajustable manualmente de 30 a 100%.

Tipos de controle:

- 3 ajustes de frecuencias de dosificación constantes manualmente: 100%, 50%, 25%;
- Entrada de contacto libre de potencial y 50%, 100% de frecuencias constantes;
- Entrada de 4-20 mA y 50%, 100% de frecuencias constantes.

Áreas de aplicación

- Cloración general;
- Aguas residuales municipales e industriales;
- Desinfección de agua;
- Biocida.

Bomba dosificadora

Datos técnicos

CONCEPT c - bomba dosificadora	
Caudal	0,72 – 24 l/h
Contrapresión:	16 a 1,5 bar
Materiales de construcción:	Noryl
	Diafragma: PTFE
	Válvulas: PTF
Potencia disipada:	12W / 16W / 24W
Alimentación eléctrica:	220V / 60Hz
Precisión de dosificación:	±0,5% en el rango de 0-100% de ajuste de la longitud de carrera
Ajuste longitud de carrera:	Manual
Ajuste remoto de dosificación:	Por pulso o vía 4-20mA
Máx. temperatura del producto dosificado:	-10 a 45°C
Tamaño máximo de sólidos:	N/A
Viscosidad máxima:	1.000mPas
Intervalo de mantenimiento:	5000 horas de operación aproximadamente
Test de fábrica:	Todas las bombas se prueban en el caudal y presión máximas
Máxima presión de succión:	50% de la presión máxima de la bomba

Materiales en contacto con el producto químico

	Cabezal de dosificación	Succión/Contrapresión	Alojamiento	Embalaje	Esferas
PVT	PVDF	PVDF	VITON / PTFE	PTFE	Cerámica

ProMinent®

Para más informaciones:

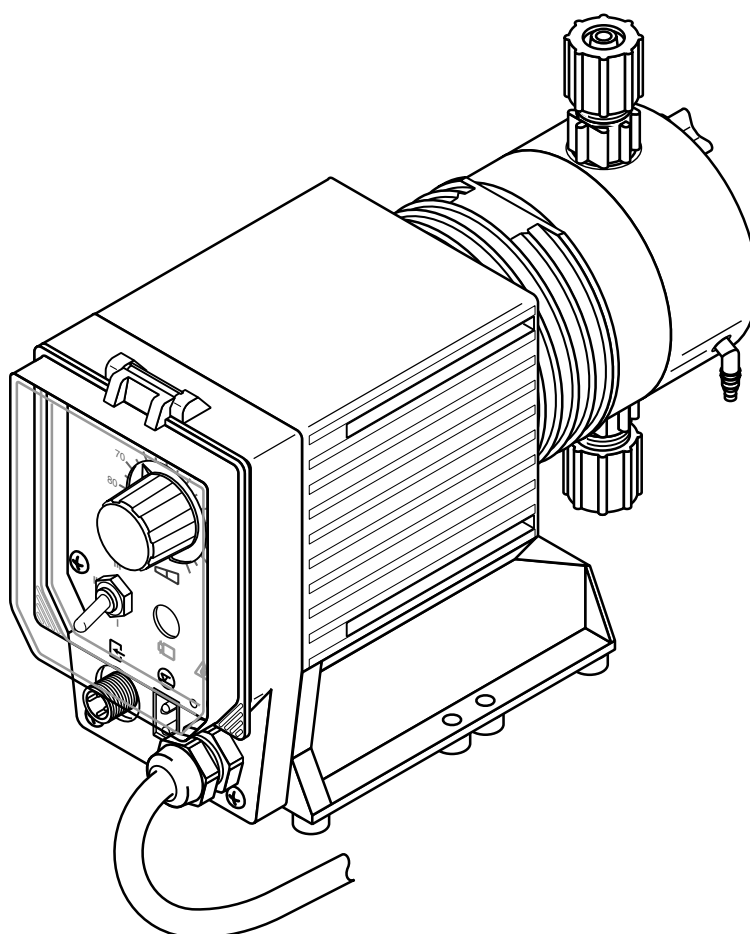
info@prominent.com

www.prominent.com

Operating Instructions

Metering Pump

ProMinent® CONCEPT c



Affix type identification plate here!

Please completely read through these operating instructions first! • Do not discard!
The warranty shall be invalidated by damage caused by operating errors!

Key to Devices and Components

Detailed description of parts on Pages:

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2	Stroke Length Adjustment Knob	12, 21, 23–25
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4	Mains Power Connection.....	12, 13
5	Connection Socket for Level Switch	11, 13
6	Connection Socket for External Activation	11, 13, 21
7	Transparent Cover	
8	Housing	23, 24
9	Head Valve	11, 12
	<u>for PP and NP Versions only</u>	
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10 a	Precision Bleeder Screw	
10 b	Palm Grip	
11	Bypass Hose Socket	11
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13	End Disc	23, 24
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Imprint

Operating Instructions CONCEPT c
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Subject to modifications

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1 Foreword

The ProMinent® CONCEPT is a microprocessor-controlled short-stroke solenoid-operated diaphragm-type metering pump for metering liquid, non-combustible media.

2 Functions

2.1 Brief Description of Functions

Metering is based on a pulsating action. For each pulse coming from the electronic control, the solenoid energizes and completes a stroke of max. 1.25 mm. The diaphragm, stabilized by a steel core, displaces the medium in the liquid end, the balls on the intake side are pressed against their seat while the balls on the delivery side open, thus releasing the medium for metering. The solenoid is deactivated on completion of the stroke and returned to its initial position by a spring. In so doing, the valve on the delivery end closes while the valve on the intake side opens so as to draw medium into the liquid end. The safety diaphragm protects the leadthrough in the housing from dust and moisture.

The solenoid-operated, diaphragm-type metering pump covers a capacity range from approx. 0.9 l/h to 23 l/h (litre specifications at max. stroke length and max. stroke rate) at a max. backpressure of 1.5 bar to 16 bar. The delivery capacity can be adjusted on the basis of the stroke length by means of a rotary knob and the stroke rate by means of a switch. The various operating statuses are indicated by an LED for operation and empty/fault signal.

The pump features a connection facility for a single-stage level switch. In addition to the internal mode of operation, external activation is also possible via floating contacts.

3 Technical Data

3.1 General Data

<i>Metering deviations</i>	All material versions –5 % to +15 %
<i>Metering reproducibility</i>	When used in accordance with operating instructions ±2 %
<i>Permissible ambient temperature</i>	-10 °C to +45 °C
<i>Electrical connection</i>	230 V, 50/60 Hz, +6 %, –10 % 115 V, 50/60 Hz, +6 %, –10 %
<i>Mean power intake</i>	Refer to type identification plate
<i>Max. current consumption</i>	Refer to type identification plate
<i>Type of enclosure</i>	IP 65
<i>Insulation class</i>	F

Technical Data

External contact activation:

Necessary contact duration 20 ms
Contact load 5 V; 0.5 mA

Contact load

Fault indicating relay (option) 250 V; 2 A

Contact load

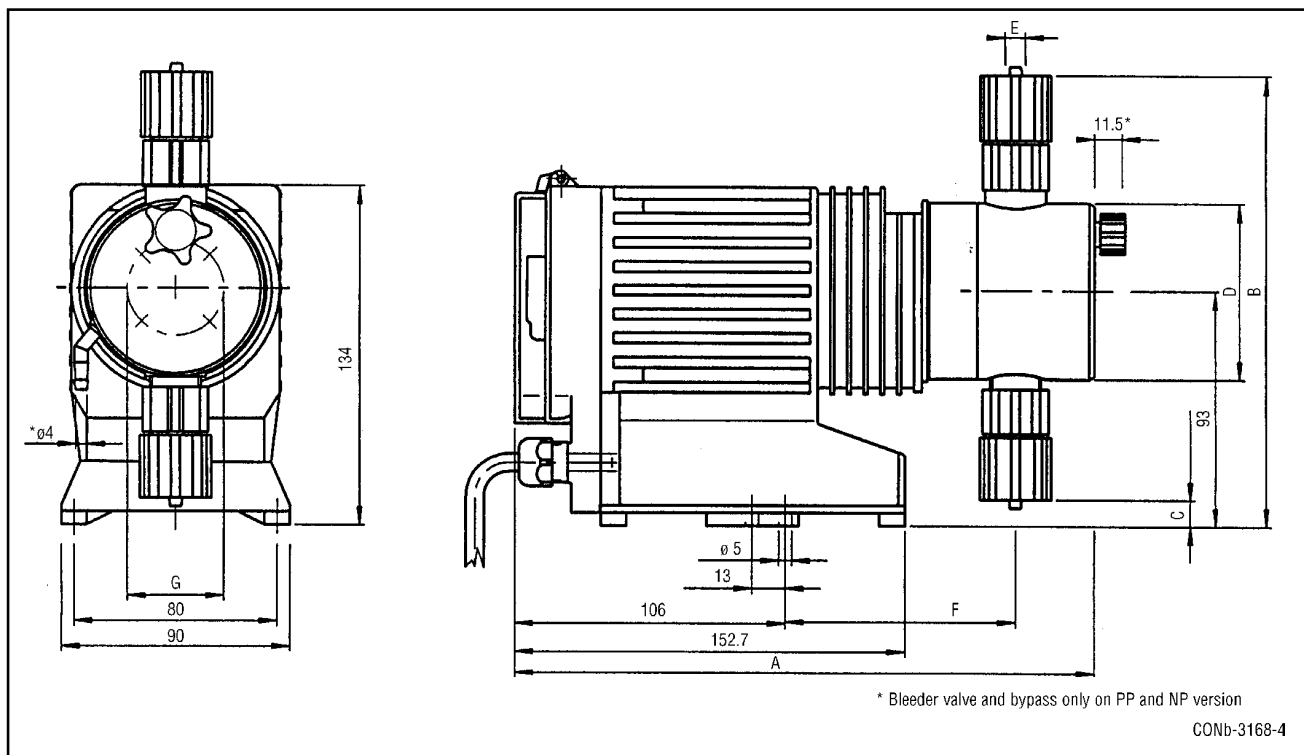
Clock generator relay (option) 42 V; 0.1 A

3.2 Capacity Data

Pump type						
Pump type CONb with 120 strokes/minute	1601	1201	0803	1002	0308	0215
Max. capacity (l/h)	1.0	1.7	3.4	2.3	8.4	14.8
Capacity (ml/stroke)	0.14	0.24	0.48	0.32	1.17	2.05
At max. backpressure (bar)	16	12	8	10	3	1.5
Max. capacity (l/h)	1.2	1.9	3.8	2.6	9.1	15.80
Capacity (ml/stroke)	0.17	0.27	0.53	0.36	1.23	2.20
At mean backpressure (bar)	8.0	6.0	4.0	5.0	1.5	1.0
Stroke rate (strokes/min)	120	120	120	120	120	120
Connection size $\varnothing \varnothing \times i \varnothing$ (mm)	6x4	6x4	6x4	8x5	8x5	12x9
Suction head (m water column)	6.0	6.0	3.0	6.0	6.0	1.5
Intake head (m water column)	1.8	2.5	2.8	2.0	1.8	1.5
Permissible admission pressure intake side (bar)	8.0	5.5	3.0	3.0	1.5	0.8
Pump type CONb with 180 strokes/min	1602	1203	0806	1003	0313	0223
Max. capacity (l/h)	1.5	2.5	5.3	3.4	13.0	23.0
Capacity (ml/stroke)	0.14	0.24	0.48	0.31	1.20	2.13
At max. backpressure (bar)	16	12	8	10	3	1.5
Max. capacity (l/h)	1.8	2.9	5.6	3.7	15.5	25.5
Capacity (ml/stroke)	0.17	0.27	0.53	0.34	1.43	2.36
At mean backpressure (bar)	8.0	6.0	4.0	5.0	1.5	1.0
Stroke rate (strokes/min)	180	180	180	180	180	180
Connection size $\varnothing \varnothing \times i \varnothing$ (mm)	6x4	6x4	6x4	8x5	8x5	12x9
Suction head (m water column)	6.0	6.0	3.0	6.0	6.0	1.5
Intake head (m water column)	1.8	2.5	2.8	2.0	1.8	1.5
Permissible admission pressure intake side (bar)	8.0	5.5	3.0	3.0	1.5	0.8

Installation Dimensions

4 Installation Dimensions



CONc

1601, 1201
1602, 1203

PP
NP

0803, 0806

PP
NP

1002, 1003

PP
NP

0308, 0313

PP
NP

0215, 0223

PP
NP

A	B	C	D	E	F	G
225	180	11	70	6x4	88	38
223	173	13	70	6x4	88	38
225	180	11	70	6x4	88	38
223	173	13	70	6x4	88	38
218	180	11	70	8x5	83	50
216	181	5	85	8x5	83	50
218	180	11	70	8x5	82	50
216	181	5	85	8x5	82	50
218	191	0	90	12x9	83	66
216	189	-3	100	12x9	83	66

Tests

EC Declaration of Conformity

We,

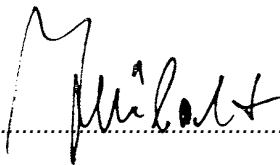
**ProMinent Dosiertechnik GmbH
Im Schuhmachergewann 5 - 11
D - 69123 Heidelberg**

hereby declare that, on the basis of its functional concept and design and in the version brought into circulation by us, the product specified in the following complies with the relevant, fundamental safety and health stipulations laid down by EC regulations.

Any modification to the product not approved by us will invalidate this declaration.

Product description:	Metering pump, Series CONCEPT
Product type:	CONc.... ,
Serial number:	see type identification plate overleaf and on device
Relevant EC regulations:	EC-machine regulation (89/392/EEC) subsequently 93/44/EEC EC-low voltage regulation (73/23/EEC) EC-EMC-regulation 89/336/EEC subsequently 92/31/EEC
Harmonised standards used, in particular:	EN 292-1 , EN 292-2 , EN 809 EN 60335-1 A6 , EN 60335-2-41 EN 50081-1/2 , EN 50082-1/2 , EN 55014 EN 60555-2 , EN 60555-3
National standards and technical specifications used, in particular:	DIN VDE 0700 T1 DIN VDE 0700 T41 DIN VDE 0700 T500 CSA Standard C 22.2 No. 0-M 91 (115 V-Version) CSA Standard C 22.2 No. 108-M 89 (115 V-Version)

Date/manufacture's signature: 28.03.1996



The undersigned: *Mister Manfred Hüholt, factory manager*

Safety Notes

6 Safety Notes



ProMinent® metering pumps must not be assembled with parts which have not been tested and recommended by ProMinent otherwise this can lead to injury to persons and damage to property for which no liability will be accepted!



Pumps must be accessible at all times to facilitate operation and maintenance. Access points must not be obstructed or blocked!



When metering hazardous or unknown liquids, always wear safety clothing (safety goggles, gloves) when working on the liquid end!



Before carrying out any maintenance and repair work always drain off and flush out the liquid end first if hazardous or unknown metered media are used! Observe data sheets of metered liquids!



The metering pump can contain water residue in the liquid end as the result of testing at the factory! If handling media which must not come in contact with water, all traces of water must be removed from the liquid end before start-up! For this purpose, turn the pump through 180° and drain off the liquid end then flush with a suitable medium from above via the intake connection.



Always depressurize the delivery line first before carrying out any work on the pump! Always discharge and flush liquid end! Observe safety data sheets for metered liquid!



WARNING:

When operating the metering pump against a closed shut-off element on the delivery side, the resulting pressure buildup can reach a multiple of the maximum permissible backpressure!

This can cause the delivery line to burst!

To avoid this, it is advisable to install a ProMinent multifunction valve which limits the maximum pressure which can be reached!

The stroke length should only be set with the pump in operation – when the load on the metering stroke setting pin is relieved temporarily – only use the clamping rings and hose sockets intended for the relevant hose diameter as well as original hoses with the specified hose size and wall thicknesses otherwise the stability and durability of the connection will not be guaranteed!

Avoid reducing hose sizes!

The next higher line cross-section or a pulsation damper should be used for longer lines and high-viscosity media.

Safety Note/Installation



DANGER:

Only qualified electricians are permitted to carry out electrical repairs (safety regulations of Employer's Liability Insurance Association: VBG 4 and ZH 1/11)!



DANGER:

Disconnect power plug or power supply line before opening the pump! Isolate relay option if applicable! Check to ensure power is disconnected! Secure pump while carrying out repairs to ensure it cannot be switched on unintentionally!

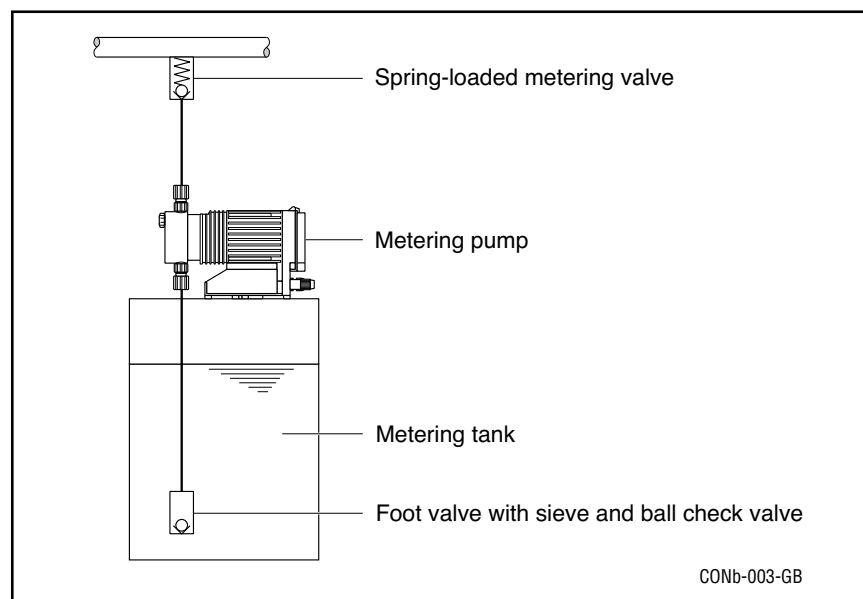


WARNING:

Only specially trained and authorized personnel are permitted to carry out repairs on metering pumps!

7 Installation

7.1 Pump Installation

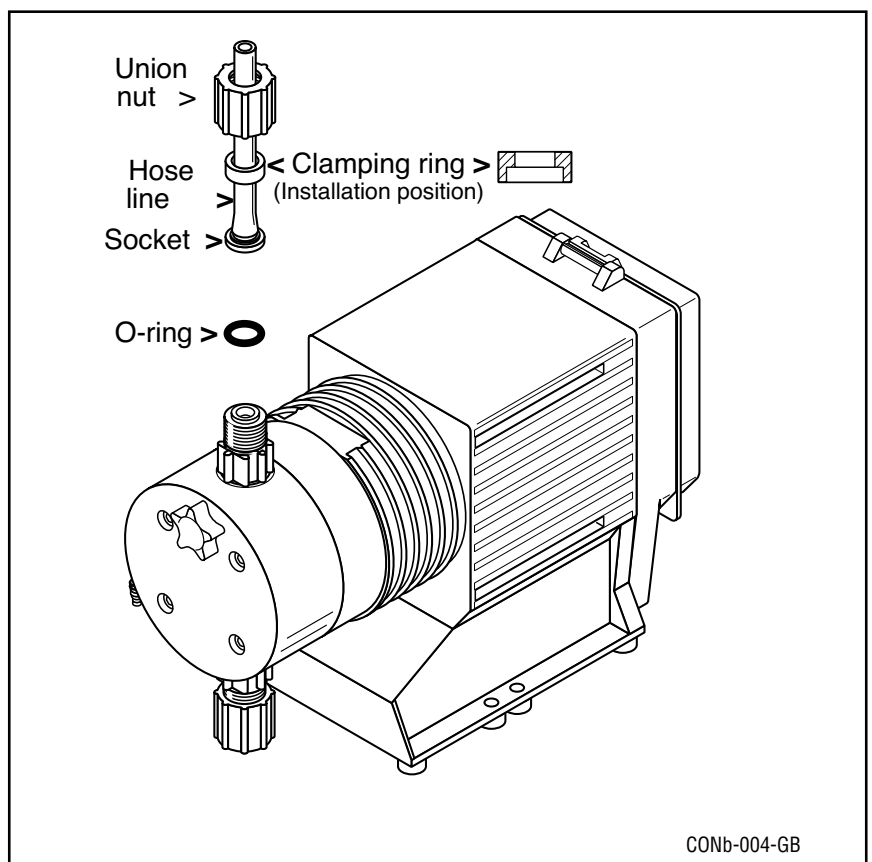


- Mount metering pump on a tank or support bracket with bolts and U-washers (Ø 5 mm).
- To ensure efficient operation, the valves of the liquid end must always be positioned vertically.
- Shorten intake lines such that the foot valve hangs approx. 10–30 mm (approx. 50 mm for heavily soiled metering solutions) over the bottom of the tank.
- Install delivery line and metering valve, pressure retention valve for free outlet.

Installation

7.2 Hose Connections

- Fit union nut and clamping ring over hose line.
- Push the hose end, cut at right angles, onto socket as far as it will go.
- Slightly widen hose end if necessary.
- Shorten hose end at right angles by approx. 10 mm for connecting several times.
- Press on hose and firmly tighten union nut.



7.3 Installation of Bypass Bleeder Line

Liquid end with bleeder valve

The bleeder valve (**10**) with bypass (**11**) is provided on the liquid end in the versions NP and PP up to Type 0417.

- Fit 4 mm inside Ø (max. 6 mm) hose on to bypass hose socket, preferably use 6 x 4 mm soft PVC.
- If PE lines are used, secure with a cable tie to prevent slipping.
- Insert free end of hose line in metering tank.
- Connect delivery line directly to delivery connection and metering valve.

Installation

7.4 Electrical Connections

7.4.1 Connection of Control Cable and Level Switch

- If applicable, connect level switch to connection socket **(5)** on the pump.
- If provided, connect contact/control cable or universal control cable to the pump connection socket **(6)**.

2-core contact control cable

If the 2-core contact control cable is connected, the pump is switched over to external mode by the pressures "1–4" in the clutch. In addition, the frequency selector switch **(1)** must be set to position "1" .

4-core universal control cable

If the 4-core universal control cable is used, the brown and black cores for changeover to external mode should be connected by wire jumpers. If necessary, the pump can be stopped potential-free in internal mode with the aid of this circuit or the frequency changeover (blue core).

7.4.2 Mains Power Connection

- Connect mains power plug in the socket provided

ATTENTION:

Take particular care to ensure that the mains voltage agrees with ratings on the type identification plate. A separate switching contact, e.g. relay or contactor should be provided for parallel connection with inductive loads.

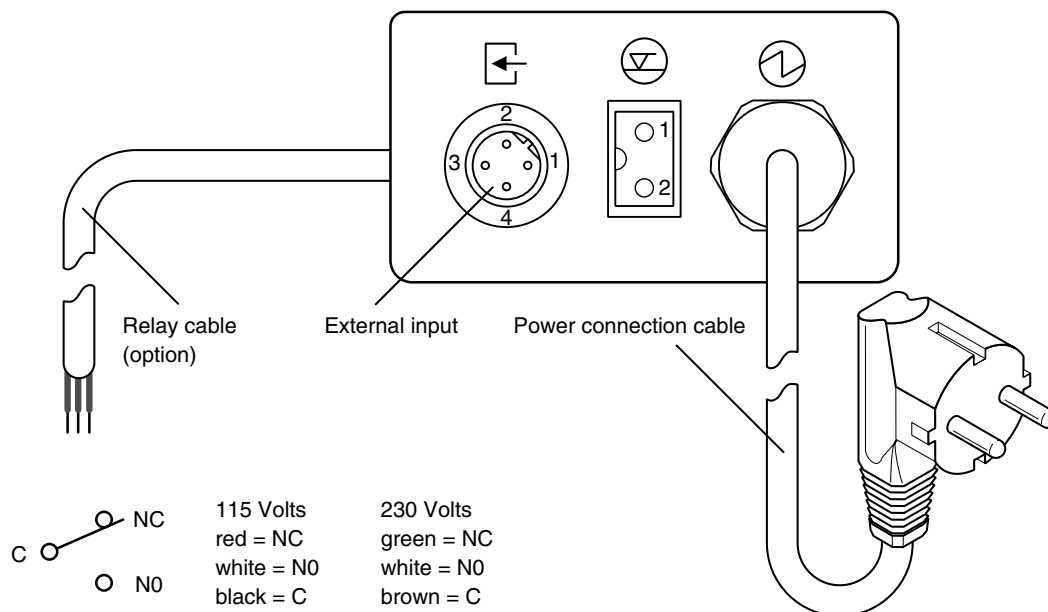
7.5 Start-Up

- In order to prime the metering pump, if necessary, remove external cable, set stroke length knob **(2)** to 100 % and stroke rate setting knob **(1)** to III.
- Open bleeder valve by approx. 1 turn.
- Allow pump to run until a little of the chemical is visible in the delivery line or emerges at the bleeder valve.
- On metering pumps without a bleeder valve, the delivery line must be released at the liquid end or a multifunction valve should be used.
- After priming, close bleeder valve **(10)** or resecure delivery line at metering valve.

Installation

7.6 Wiring Diagram

View of connector block on pump



Socket assignments:

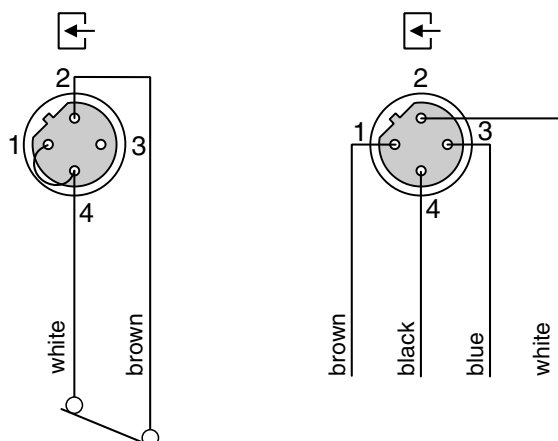
External activation :

- 1 = External switchover (+5 V)
- 2 = External contact (+5 V)
- 3 = Floating frequency switchover to F3
- 4 = GND

Level switch :

- 1 = End cutout (+5 V)
- 2 = GND

View of cable sockets on control cable (2-pin or 4-pin) or on level switch connection cable



External/contact cable:

Close contact -
metering stroke

Universal control cable:

External switchover: 1 brown + 4 black/GND
brown + black connected = Pump in external mode, brown + black open = Pump meters in internal mode

External/contact: 2 white + 4 black/GND
Close contact - metering stroke (1 and 4 connected by jumper)

Frequency changeover: 3 blue + 4 black/GND
Close contact - pump operates with F3

CONb-005-GB

Installation

7.7 Determining Delivery Capacity by Means of Diagrams

7.7.1 General

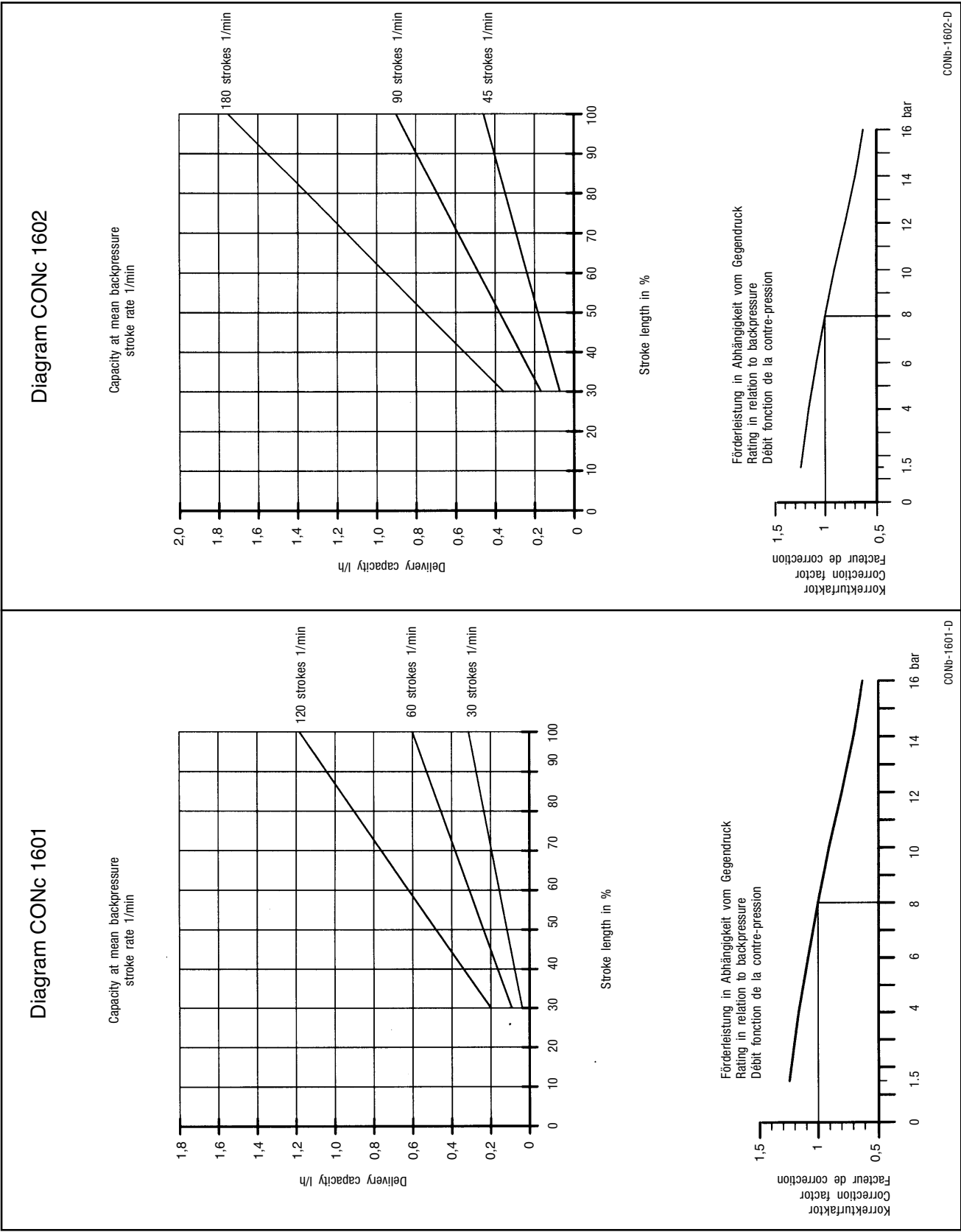
1. Determine required delivery capacity.
2. Determine correction factor in accordance with the diagram "Capacity as a Function of Backpressure".
3. Divide required delivery capacity by correction factor.
4. Find calculated value on the vertical scale in the diagram "Capacity at Mean Backpressure" and draw a horizontal line from this point over the entire diagram.
5. Draw a vertical line from the point of intersection with the frequency characteristic curve down to the stroke length scale. Read off the stroke length.
6. Set determined stroke length on the pump.
7. Set pump frequency rate on the pump as indicated at the point of intersection on the frequency curve.
8. If the horizontal line intersects two frequency characteristic curves starting from the capacity scale in the diagram "Capacity at Mean Backpressure", if there are no other compelling reasons, the lower frequency should be selected for the sake of metering accuracy.
9. The result obtained in this way is a reference value. To obtain accurate metering, the delivery capacity must be monitored and corrected if necessary.
10. Refer to Technical Data for metering accuracy.
11. The capacity data are referred to water, the correction factor was determined at a stroke length of 70 %.

Example: Metering pump Type 1201

- Required metering capacity 0.7 l/h at 8 bar
- Correction factor at 8 bar in accordance with diagram = 0.85
Delivery capacity for diagram = $0.7 \text{ l/h} / 0.85 = 0.82$
- The horizontal line drawn at 0.82 l/h intersects two frequency characteristic curves. The frequency curve which results in the highest stroke length should be selected. Here in this example this is the frequency characteristic curve with 60 strokes/min. Set this frequency on the pump.
- The value obtained for the stroke length is 92 %.
- Diagrams for examples (refer to Pages 15 to 20).

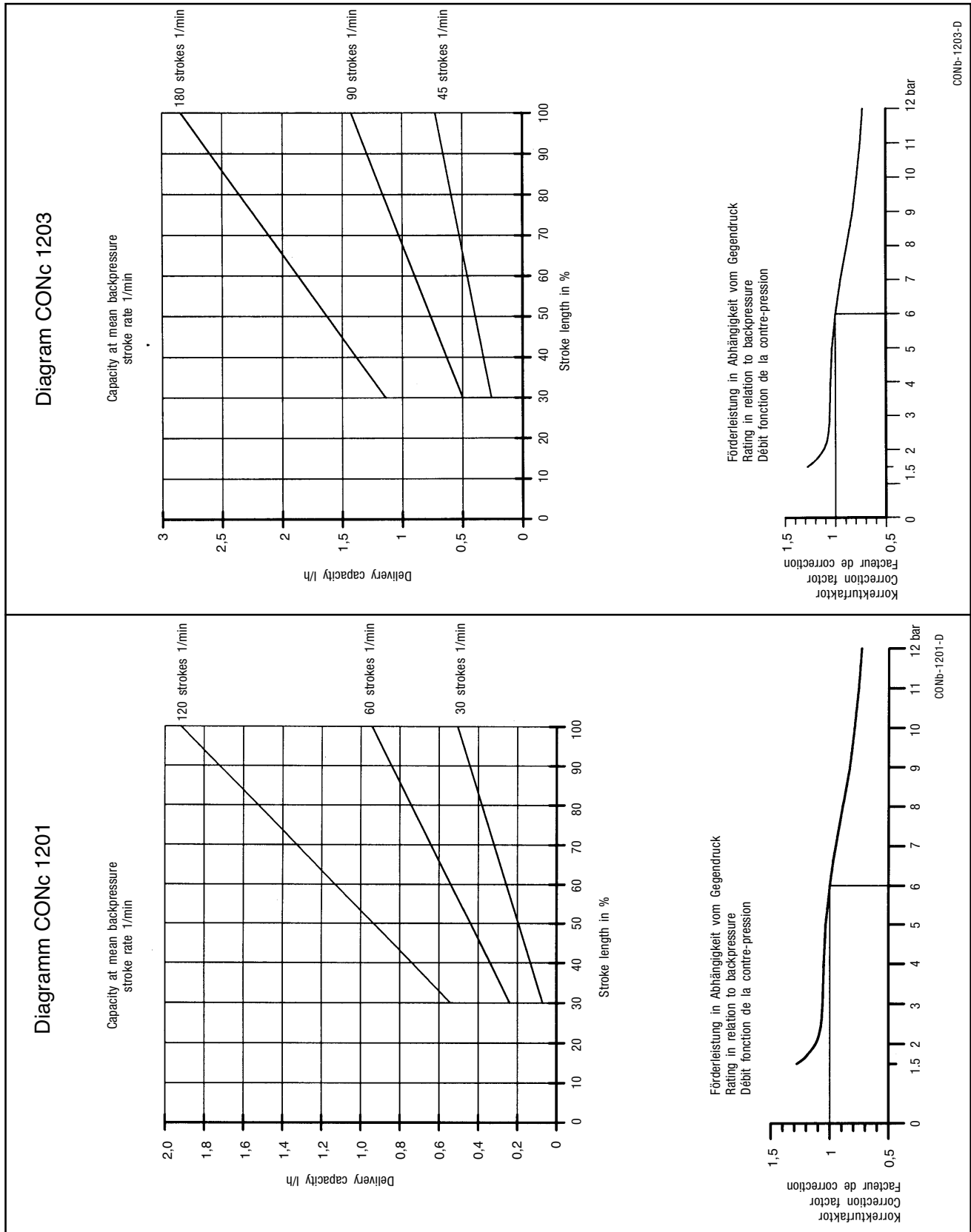
Installation

7.7.2 Diagram 1601, 1602



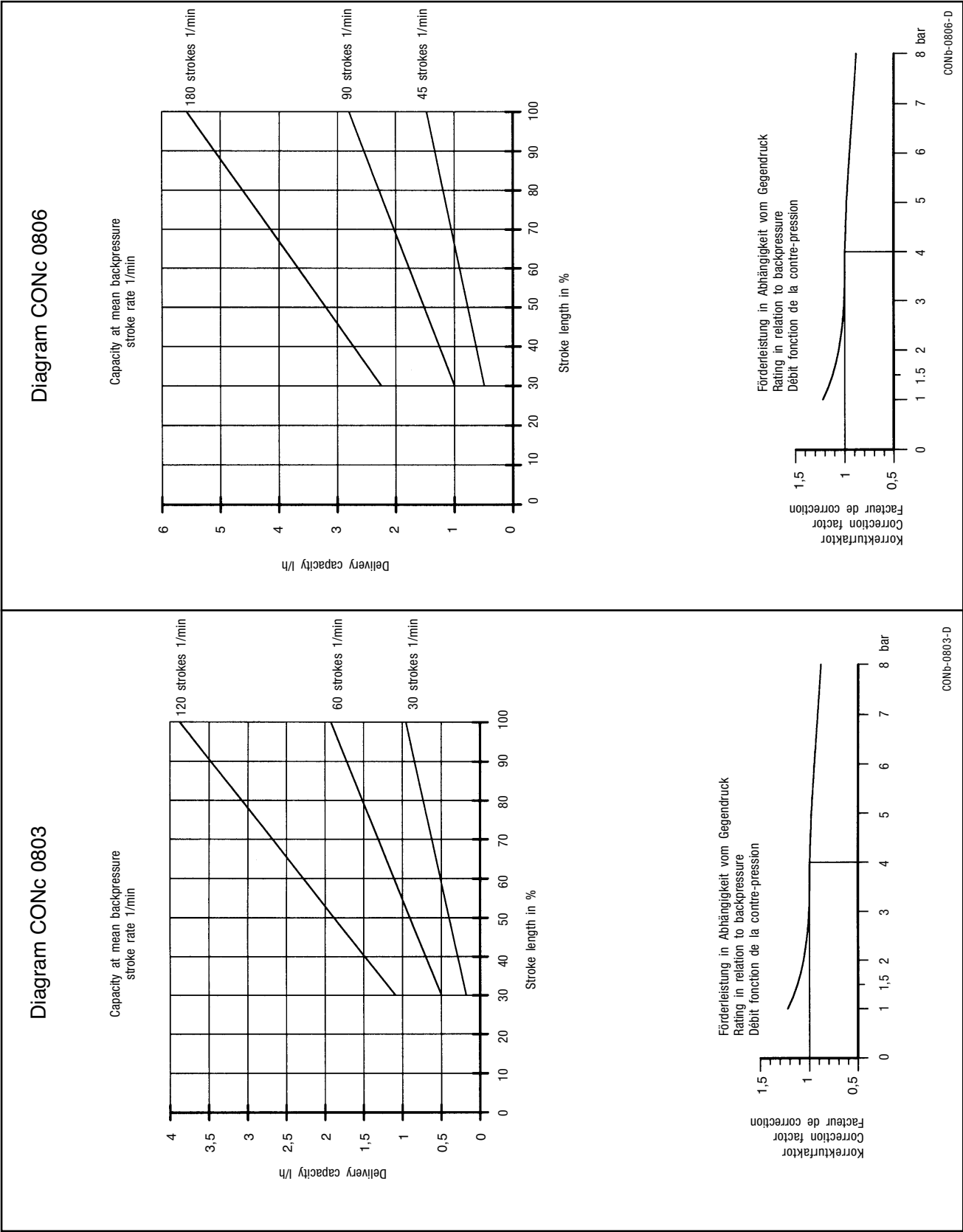
Installation

7.7.3 Diagram 1201, 1203



Installation

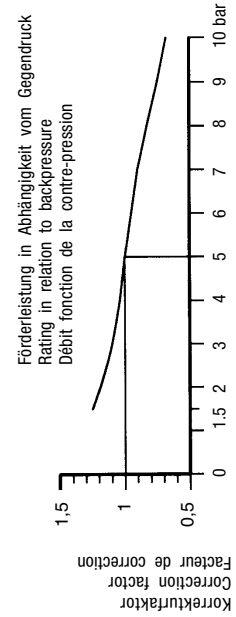
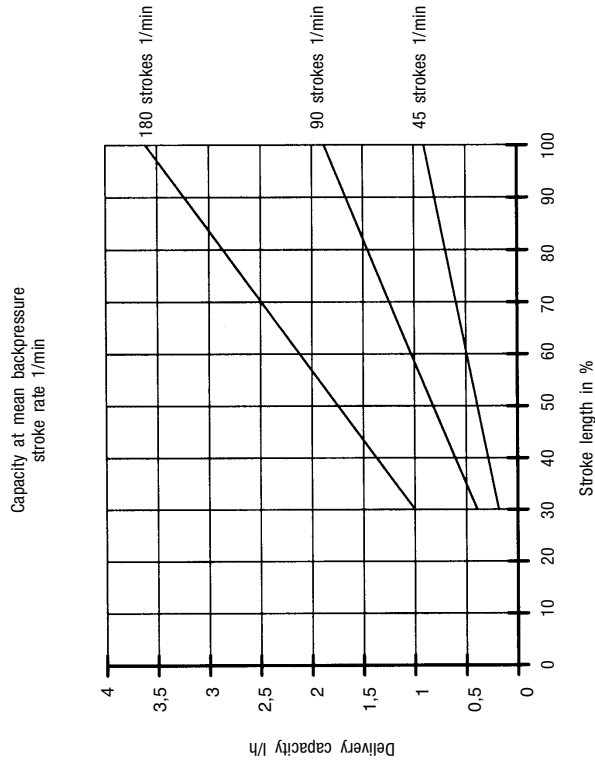
7.7.4 Diagram 0803, 0806



Installation

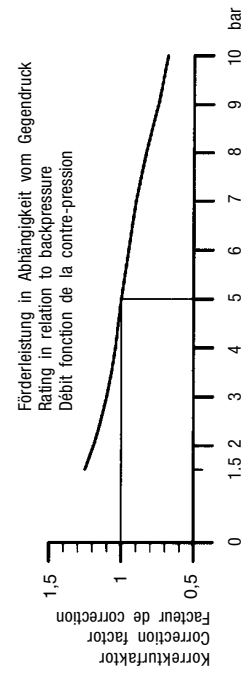
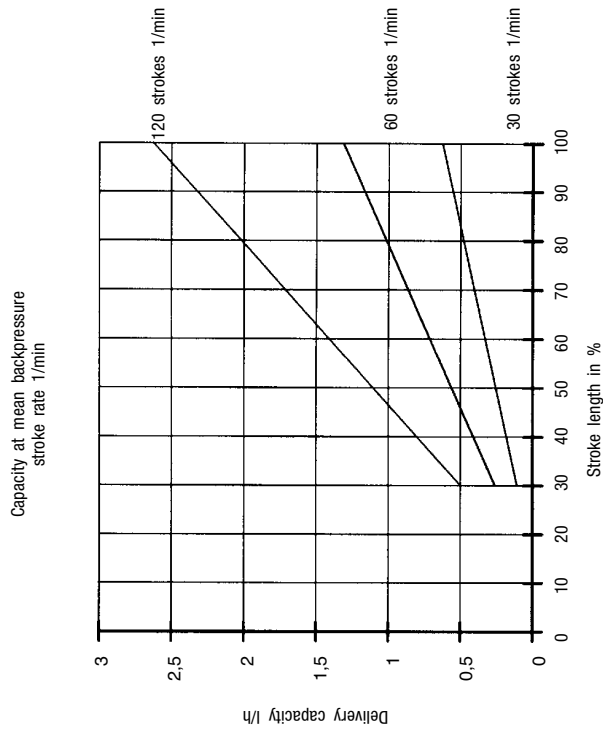
7.7.5 Diagram 1002, 1003

Diagram CONc 1003



CONb-1003-D

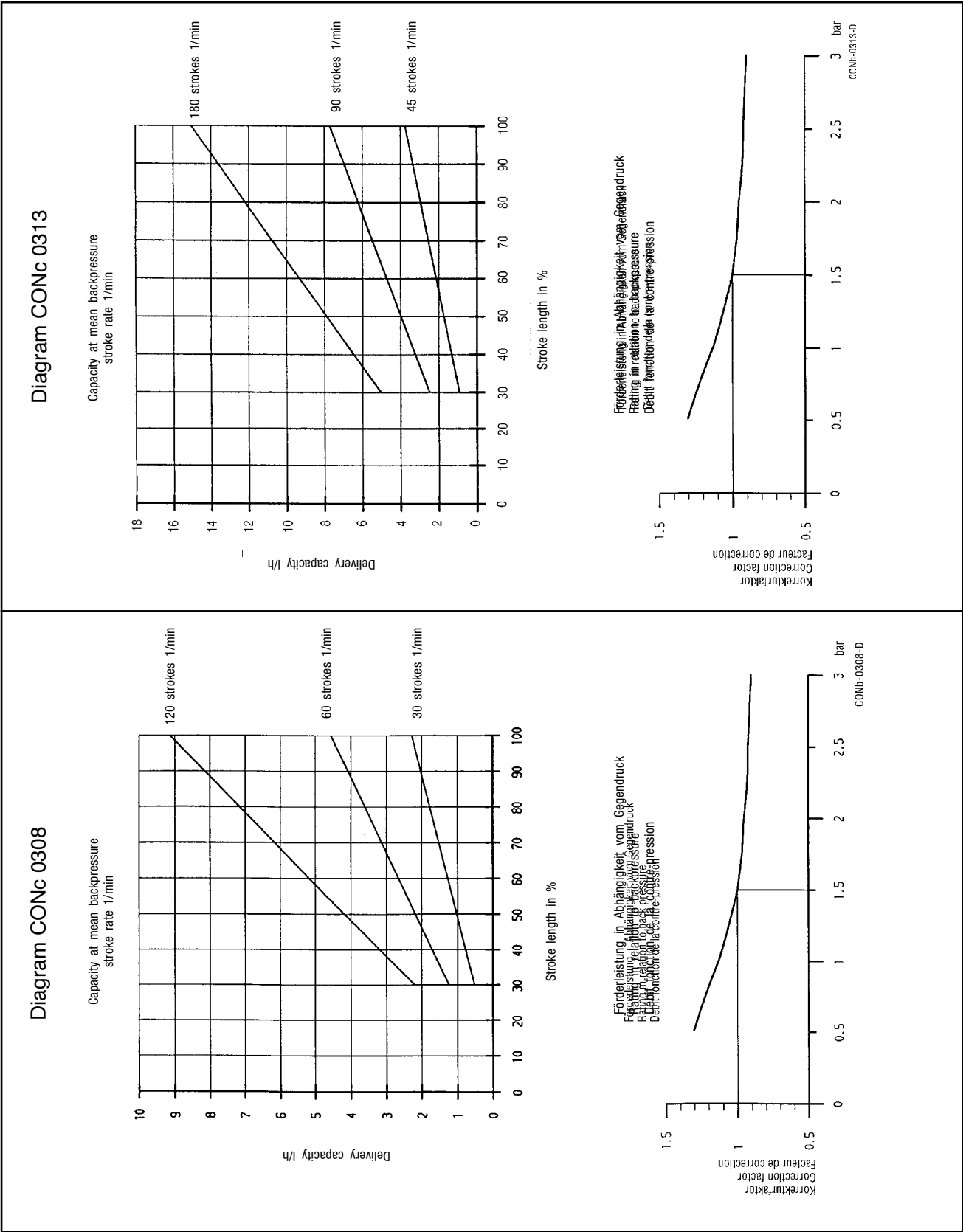
Diagram CONc 1002



CONb-1002-D

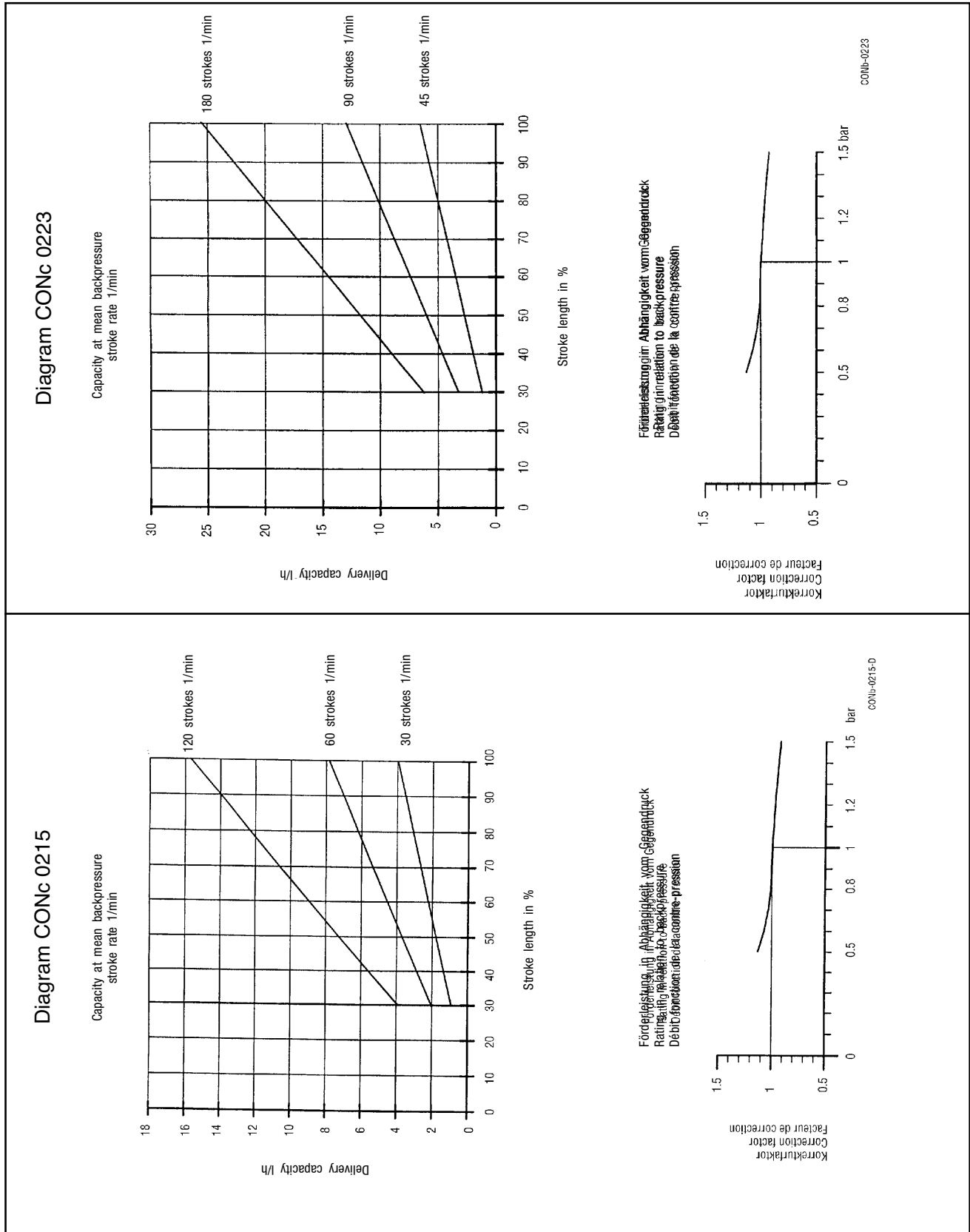
Installation

7.7.6 Diagram 0308, 0313



Installation

7.7.7 Diagram 0215, 0223



Operation/Maintenance

8 Operation and Setting

8.1 Stroke Length Setting

This setting is determined by means of a diagram (refer to Page 15 to 20) and set with the stroke length setting knob **(2)**.

8.2 Pump Frequencies (Rates)

The stroke rate is set by means of the frequency selector switch **(1)**:

Setting III corresponds to 100 %, setting II 50 % and setting I, depending on type of control, 25 % or 0 % stroke rate

Control type	III	II	I	External
0	120/180	60/90	30/45	–
1	120/180	60/90	0	+

8.3 What is the Result of Incorrect Operation?

The control cable must be disconnected when priming externally activated pumps. Overmetering may occur if this cable is not reconnected after priming. This could lead to serious damage in certain applications, e.g. disinfection of drinking water with chlorine bleaching agent.

The LED **(3)** lights orange if an illogical setting is made on the pump, e.g. external connector **(6)** is plugged in and switch **(1)** is set in position II or III.

9 Maintenance

Maintenance of the CONCEPT metering pump is restricted to checking

- the liquid end screws (firm fit)
- the delivery lines (firm fit)
- the head and intake valves (firm fit)
- the leakage hole at the end disc (moisture can indicate diaphragm failure)

Troubleshooting

10 Troubleshooting

Breakdown assistance

In order to be able to repair a small breakdown at any time, we recommend to always keep a spare parts kit in stock - consisting of diaphragm, valves, balls and seals. See Page 26 for order information.

No pump intake despite full stroke movement and venting

Possible cause:

Crystalline deposits caused by valves drying out

Remedy:

- Remove intake hose out of metering tank and thoroughly flush out liquid end
- If this provides no improvement, remove and clean valves
- The valve seat in the intake and head valve can be pressed out with a 3 mm $\varnothing$ drift.

Warning lamp (3) lights red

Possible cause:

Chemical deficiency

Remedy:

- The fault signal is acknowledged automatically by filling the supply tank

Warning lamp (3) lights orange

Possible cause:

- External cable connected and frequency selector switch not set to position I.

Remedy:

- If external mode is required, set frequency selector switch to position I.
- If internal mode is required, disconnect external control cable.

Liquid emerges at head end

Possible cause:

Delivery unit leaking at metering diaphragm

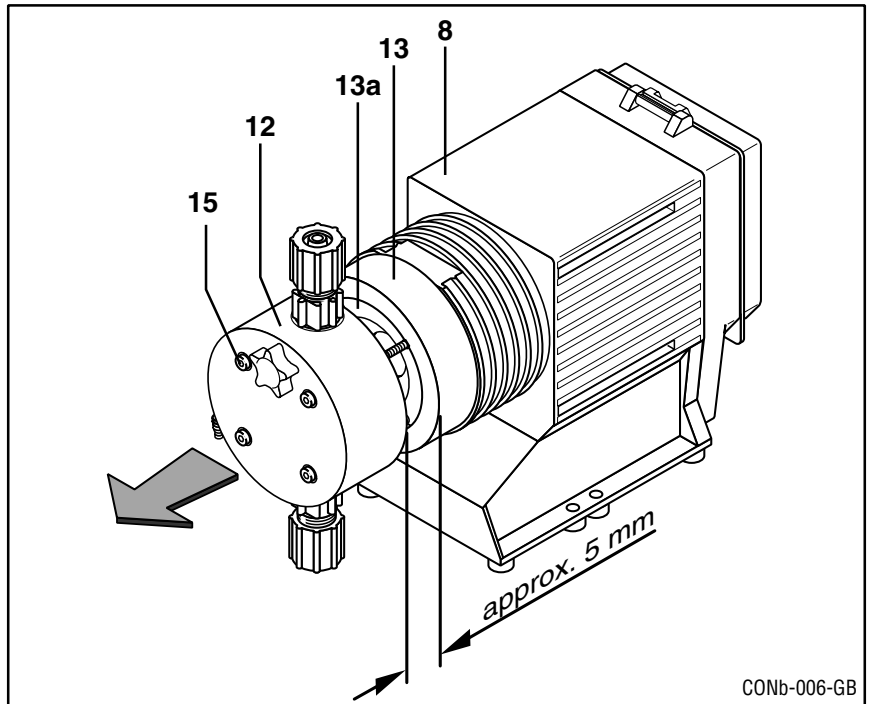
Remedy:

- Retighten screws in liquid end crosswise.
- If liquid still leaks out this indicates the diaphragm has failed and must be replaced.

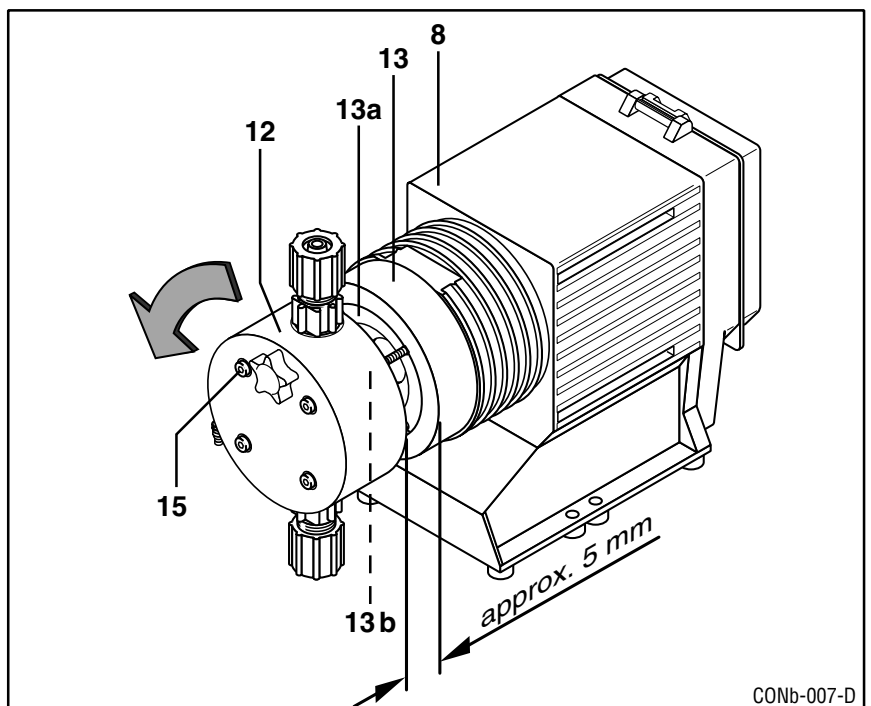
Troubleshooting

Replacing metering diaphragm

- With pump in operation, set stroke length to "0" with rotary knob (2)
- Release screws (15)

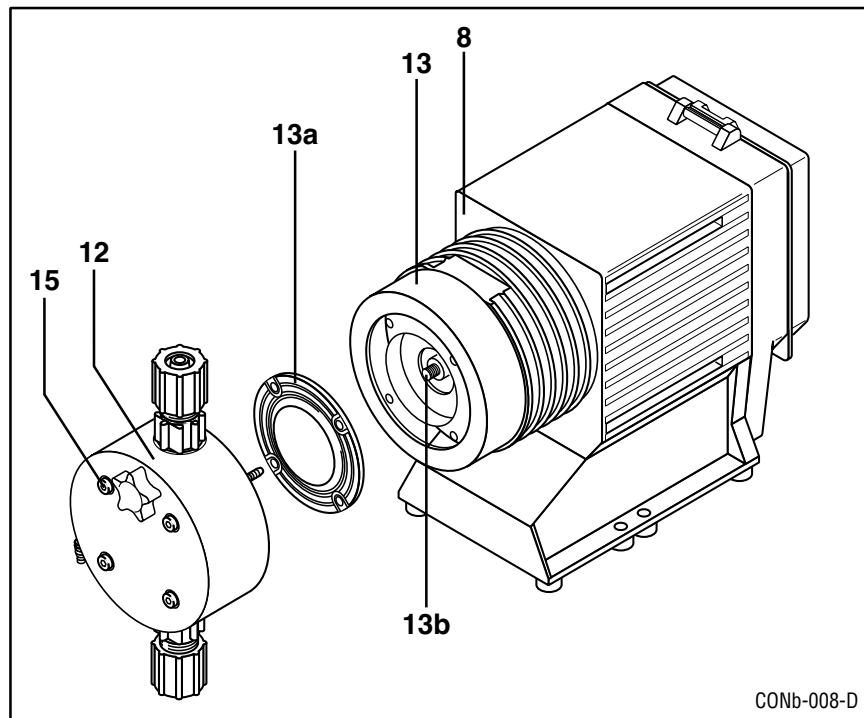


- Pull liquid end (12) together with screws out of end disc (13) and pump housing (8) (approx. 5 mm) until the screws no longer offer any resistance when turning the liquid end.
- Hold housing (8) in the left hand and release diaphragm (13a) with the right hand by lightly jolting liquid end (12) together with end disc (13) in counterclockwise direction and release screws (15) from drive spindle (13b).



Troubleshooting

- Now pull liquid end together with screws out of diaphragm and unscrew the entire assembly from the drive spindle.
- Remove diaphragm (13a) from end disc (13).



- Screw on new diaphragm (13a) as far as it will go on drive spindle (13b) and check thread movement.
- Once again remove diaphragm from drive spindle.
- Fit end disc on housing.
- Insert diaphragm in end disc and screw on by 2 thread turn

ATTENTION:

Turn diaphragm such that the 4 holes in the diaphragm and end disc are precisely aligned! The outlet in the end disc must face downward!

- Fit liquid end (intake connection must face downward!) with screws on to diaphragm and end disc (once again leave approx. 5 mm space between liquid end and end disc) such that the parts can still be turned.
- Now slowly turn parts in clockwise direction until diaphragm is firmly fitted (resistance of return spring can be felt)

ATTENTION:

Do not overturn diaphragm, particularly in Type 1601!

Troubleshooting/Repair

- Set stroke length to 100 % with rotary knob **(2)** and, with metering pump running, slowly turn the complete delivery unit in clockwise direction until the intake connection faces vertically downward.
- Stop pump.
- Tighten 4 screws **(15)** crosswise.

ATTENTION:

Observe screw tightening torques!

Tightening torque for screws

*M4: **2.5 to 3 Nm***

*M5: **4.5 to 5 Nm***

NOTE:

Check the tightening torque of the liquid end screws after 24 hours of operation!

Additionally check the tightening torques on PP pumps every 3 months!

- Set required stroke length with pump in operation.

11 Repair



Return the metering pump for repair only in a clean condition with the liquid end flushed clean.

Pumps which are used for metering radioactive media must not be shipped!

Spare Parts

12 Spare Parts

In order to repair small breakdowns at any time we recommend to always keep a spare parts kit in stock. This kit contains the following wearing or spare parts:

- 1 Intake valve, complete
- 1 Head valve, complete
- 1 Set of seals
- 1 Metering diaphragm
- 1 Set of connection parts

Additionally for NS 3 (self-bleeding) version:

- 1 Bleeder valve, complete
- 1 Connection for delivery side or bypass

Spare parts kit for

Pump type:	Order number
1601/1602 PP1	74.03.61.1
1201/1203 PP1	74.03.80.1
0803/0806 PP1	74.03.84.3
1002/1003 PP1	74.03.88.4
0308/0313 PP1	74.04.97.3
0215/0223 PP1	74.04.99.9
1601/1602 NP6	74.05.51.7
1201/1203 NP6	74.05.52.5
0803/0806 NP6	74.05.53.3
1002/1003 NP6	74.05.54.1
0308/0313 NP6	74.05.55.8
0215/0223 NP6	74.05.56.6
1601/1602 NS3	79.20.33.3
1201/1203 NS3	79.20.34.1
0803/0806 NS3	79.20.35.8
1002/1003 NS3	79.20.36.6

Notes
